

Pavan Kumar Ambekar

Penang, Malaysia

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SUMMARY

Senior DFT Engineer with 5+ years of experience in advanced-node ASIC development (2nm/5nm), specializing in scan architecture, ATPG, MBIST, and RTL-level testability improvements. Proven ability to debug complex DFT issues, optimize coverage, and drive end-to-end DFT implementation from RTL to sign-off.

EXPERIENCE

Senior DFT Engineer — Quest Global (Clients: Kinara / NXP, Socionext)

Jun 2025 – Present

Penang, Malaysia

Kinara (NXP) — Current Project

- Driving DFT implementation with RTL-level design modifications to enable testability.
- Implemented EDT and OCC insertion, optimizing compression and clock control strategies.
- Led scan insertion ensuring timing- and congestion-aware implementation.
- Developed and executed ATPG (Stuck-at and Transition Faults) with coverage improvement.
- Debugged ATPG coverage loss due to uncontrolled (UC) nets/flops and incorrect OCC placement.
- Performed RTL and Gate-Level Simulation (GLS) for validation.
- Executed formal equivalence checks (LEC).
- Collaborated with RTL and PD teams to resolve testability and timing issues.

Socionext — SHN-SNI Project (5nm)

- Owned end-to-end DFT flow from RTL validation to ATPG sign-off.
- Performed SpyGlass DFT checks to clean RTL-level violations.
- Achieved 99.6% Stuck-at and 96% At-speed coverage.
- Executed scan insertion, ATPG, and simulation (zero-delay and timing).
- Conducted DFT sign-off checks (LEC, MUSE, DFTV).

Senior DFT & Synthesis Engineer — Mirafra Technologies (MediaTek)

Jun 2023 – Jun 2025

Bangalore, India

- Worked on advanced-node (2nm) SoCs across multiple projects.
- Performed synthesis, scan insertion, and ATPG.
- Achieved 99.6% SA and 96% TDF coverage.
- Executed DFT validation, simulation, and formal checks.
- Collaborated with cross-functional teams for coverage and integration closure.

DFT Engineer — TechMahindra Cerium Systems

Apr 2022 – Jun 2023

Bangalore, India

- Led DFT implementation for CDAC Vega (14nm), including MBIST and IJTAG.
- Validated Intel Xeon (7nm) MBIST and IJTAG flows.
- Worked on GUC ASIC DFT flow validation.

DFT Engineer — Ziotron Consulting

Dec 2020 – Apr 2022

Bangalore, India

- Owned DFT for 4 blocks including scan insertion and ATPG.
- Achieved 99.1% Stuck-at and 89% At-speed coverage.
- Performed DRC cleaning, simulation, and coverage improvement.

TECHNICAL SKILLS

DFT & ASIC Flow: Scan, EDT, OCC, ATPG (S@/TDF), MBIST, IJTAG, Coverage Analysis, DFT Sign-off

Verification: RTL Simulation, GLS, LEC, SpyGlass, Formal Verification

EDA Tools: Tessent, Cadence Genus, Formal Compiler, VCS, QuestaSim, Verdi, DVE

Programming: Verilog, VHDL, PDL, Bash, Shell, HTML

EDUCATION

MSc Microelectronics — Newcastle University, UK (First Class with Merit)

BE Electronics & Communication — Siddaganga Institute of Technology, India (First Class)